

PROVISIONAL PATENT APPLICATION

Driven-Medium Metrology Instruments and Systems: Channel-Response Analyzer, Programmable Microforce Source and Force-Sensor Calibrator, Active Coupling Survey and Gradiometry System, and Coupling-Referenced Inertial-Navigation System, and Methods of Operating Them

Inventor: Christopher F. Milam

Filing type: U.S. Provisional Application under 35 U.S.C. §111(b)

This specification is complete and self-contained. Applicant's companion applications on support-force modulation (PROV-A), media and arrays (PROV-E), and rate control (PROV-G) are nonessential background only; no material essential to written description, enablement, or the claims is incorporated from them. Every example under PROPHETIC EXAMPLES is prospective; no experiment or nonzero result is represented as performed.

Portfolio prophetic-example statement: All examples identified as prophetic are prospective; no prophetic example is represented as an experiment performed or a result achieved.

FIELD

Measurement instruments and instrumented systems built around a driven ordered-medium module. More particularly: (C1) an analyzer that drives a specimen at its precession resonance, measures the signed synchronous support-force response, and classifies the response route of the driven angular momentum (the channel-response or λ -partition readout); (C2) a programmable, sign-reversible, modulated microforce source and force-sensor calibrator whose output is empirically calibrated against a separate force reference; (C3) an active coupling survey and gradiometry system in which a driven reference body chopped at a selected frequency is read by synchronous force detection along a traverse, down a borehole, or in time at a fixed observatory; and (C4) a coupling-referenced inertial-navigation and accelerometer-bias calibration system integrating the driven reference with an inertial measurement unit. Each product is disclosed together with its method of operation, its factory qualification and calibration, and the validation gates that govern what it may report.

BACKGROUND

No instrument in the extant literature reports the partition of driven angular momentum between a body's intrinsic-alignment ("twist") channel and its orbital-winding ("writhe") channel; the quantity is not named there, though its two limiting behaviors are anchored by century-old measurements (Einstein–de Haas and Barnett on the alignment side; gyroscopic response on the winding side). Those measurements establish angular-momentum exchange and rotational dynamics respectively; they do not by themselves

establish that the two routes produce opposite signs of vertical support-force response. That sign relationship is the applicant's hypothesis and is treated throughout as a quantity the analyzer measures and classifies, never as a presumed result.

Force-sensor calibration below the milligram-force class conventionally relies on deadweights, which are neither sign-reversible nor rapidly modulated, and on electrostatic, inertial, and piezoelectric references, which can be modulated but are tied to their own transducer geometry. Passive gravimeters and gradiometers read the local acceleration and its gradient with high precision but have no commandable reference: nothing in the instrument can be switched on, off, or reversed to separate a hypothesized coupling term from ordinary source terms. Lock-in detection is an established technique and is not asserted here as the novelty; the distinctive proposition is an **actively driven reference body whose support-force response to a commanded drive state is empirically mapped, sign-selectable, and modulated**, so that the response—whatever its ultimate physical account—can be read synchronously, calibrated against separate references, and used as a product output under stated validity conditions.

Hypothesis of reach (applicant's hypothesis; not relied on for utility). The applicant hypothesizes that the same unknown the C1 analyzer classifies—which channel driven angular momentum deposits into—governs the sign of driven support-force modification at bench scale and bears on published Einstein–Cartan treatments in which an intrinsic-spin census term bounds cosmological contraction (Trautman, *Nature Phys. Sci.* **242** (1973) 7). No mathematical identity connecting those scales is asserted. The utilities relied on for this application are the material-characterization, calibration, survey, and navigation utilities disclosed below, each of which is operable from the empirically calibrated response of the device regardless of the ultimate physical account of that response.

Applicant's published exhibit corpus (nonessential background). Applicant maintains a public, commit-dated exhibit series presenting the underlying research program in didactic depth (the Normal Realism / POAMS exhibits, <https://cfmilam.github.io/nr-poams-exhibits/>). Those pages are applicant's own prior publications and are cited as nonessential background teaching only; no material essential to enablement, written description, or the claims is incorporated from them.

SUMMARY

Four instrument products and their operating methods are disclosed, all built on one **driven-medium module** (FIG. 1): an ordered medium; an alignment-field source; a coupling structure and drive source applying an oscillating drive at the medium's located precession resonance; a force transducer reading the medium's support force through a defined load path; a reaction frame that carries the alignment-field source and coupling structure to a support separate from the force transducer; synchronized acquisition; a controller; and a calibration memory holding an empirical response map with provenance and validity envelope.

1. **C1 — Channel-response analyzer.** Drives the specimen at fixed alignment sense, toggles the drive, reverses alignment sense in separate settled blocks, decomposes the synchronous differential

support force into sense-odd and sense-even components, tests validity criteria, and reports an **operational response class** with uncertainty—or *unclassified*. A physical channel-partition estimate λ is reported only where independently established endpoint standards exist in the classification map.

2. **C2 — Programmable microforce source and force-sensor calibrator.** A driven module whose output force offset is commanded in sign, amplitude, waveform, and modulation frequency from a stored empirical transfer map determined against a **separate** force reference and monitored by an independent source-monitor channel; commands outside the validity envelope are inhibited or flagged.
3. **C3 — Active coupling survey and gradiometry system.** One or more driven reference units chopped at f_c under a common clock, synchronously detected, position- and time-registered, and reduced through a calibrated response map to a **normalized coupling-response coefficient** with uncertainty and validity flags; gravimetric or source interpretation is a conditional output. Gradiometer, vehicle, borehole, and observatory embodiments and an independent survey method are disclosed.
4. **C4 — Coupling-referenced inertial-navigation and bias-calibration system.** The driven reference co-mounted with an inertial measurement unit; a stored empirical observation model; an estimator that separates accelerometer bias from scale factor and from local coupling response using motion-state discrimination, orientation-reversal, drive-off zero checks, and an observability test; corrected inertial state with propagated uncertainty; explicit valid, degraded, and invalid states; operation without contemporaneous satellite navigation input for a characterized interval and error envelope.

Layered scope (FIG. 2). Every product output is disclosed at four layers: (L0) the raw synchronous observable—the support-force change synchronized with the commanded drive state; (L1) the empirically calibrated response coefficient obtained by applying a stored map with stated uncertainty; (L2) an inferred quantity (response class, λ , force offset in newtons, coupling or gravimetric coefficient, accelerometer bias) reported only under stated calibration and validity conditions; and (L3) a controller or system that uses the inferred quantity operationally. No layer presumes the sign, magnitude, additivity, or scaling of the response; the empirical map admits zero.

Three-tier product architecture (FIG. 3). (i) The ultimate product and its deployed operating method lead; (ii) factory qualification, calibration, and commissioning establish the empirical map and operating envelope; (iii) validation and interpretation gates govern what a deployed unit may report. A purchaser operates a *qualified module* from its stored map without repeating the qualification laboratory.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 Driven-medium module — common product architecture (ordered medium, alignment-field source, coupling structure and drive chain, force transducer, reaction frame and load paths, synchronized acquisition, controller, calibration memory, environmental sensors). FIG. 2 Layered scope — layers L0

through L3 from raw synchronous observable to operating system, with the validity gate between L1 and L2. FIG. 3 Qualification and validation flow — factory qualification, empirical map generation, deployed operation, runtime witnesses, recalibration. FIG. 4 C1 channel-response analyzer — product architecture and toggle/reversal timing with sense-odd/sense-even decomposition. FIG. 5 C1 classification map — response classes, endpoint standards, unclassified region, and the condition under which λ is reported. FIG. 6 C2 programmable microforce source and force-sensor calibrator — module on the sensor under calibration, independent source monitor, calibration memory, and the corrected traceability chain through a separate force reference. FIG. 7 C3 active coupling survey system — unit architecture with common-clock chopping, synchronous detection, registration, calibrated coefficient output; vehicle, borehole, and observatory embodiments. FIG. 8 C3 gradiometer — two or more units on fixed baselines, baseline-projected gradient, and the inversion conditions for source localization. FIG. 9 C4 coupling-referenced inertial-navigation system — IMU, driven reference, observation model, bias/scale/coupling estimator, motion-state discrimination, corrected state and error bounds. FIG. 10 Controller state machine — common to C1–C4: qualify, valid, degraded, out-of-envelope, invalid/unclassified, safe-off.

DEFINITIONS (the applicant acts as his own lexicographer; these definitions govern the claims)

Portfolio operating terminology. "POAMS-native" and "direct-mode" identify the prophetic operating tier that applies the disclosed POAMS bookkeeping directly rather than requiring a preexisting nonzero empirical response map. "Commanded" and "selectable" identify controller- addressable input states and do not promise a nonzero output. "Independently validated" means reproduced under the disclosed qualification protocol against a separate reference; "empirically established" means recorded with uncertainty and provenance in a stored response map. Extant-framework labels appearing in quotation marks or introduced by "extant label:" identify measured phenomena by their customary names; their use commits neither the specification nor the claims to any entity ontology historically associated with those names. Every measured claim quantity is fixed by a recited instrument reading. Hypothesis parameters are identified expressly and are not presumed to have independently measured values.

1. **Ordered medium / aligned internal-circulation population** — a body whose constituent angular momenta can be aligned by a static field, measured in aggregate as magnetization (A/m; standard magnetometry); reference medium single-crystal yttrium iron garnet (YIG).
2. **Alignment field; alignment sense** — the static bias field (oersted or tesla) applied to establish a measured alignment state of the population; "sense" is its polarity.
3. **Precession resonance** — the absorption resonance of the aligned population under the alignment field (extant label: ferromagnetic resonance, FMR), located operationally as the absorption extremum to which the drive servo locks.
4. **Oscillating drive** — a time-periodic influence applied to the body at a stated turn rate through a coupling structure (cavity, coil, or antenna), specified by its frequency, delivered power, and pulse timing at the stated port.

5. **Drive state (x)** — the commanded and independently measured operating variables for one observation: alignment-field magnitude and sense; drive frequency, phase, incident and absorbed power, pulse width, repetition rate, duty factor, and waveform; resonance mode; aligned fraction; body, mount, and ambient temperature; pressure; orientation; and time.
6. **Support force** — the force (newton) read by the force transducer supporting the body through the defined load path.
7. **Synchronous observable (L0)** — the change in support force synchronized with a commanded change of drive state. For a toggle protocol, $\Delta F_s = \text{mean}(F_{\text{ON}}) - \text{mean}(F_{\text{OFF}})$ at alignment sense s ; for a chopped protocol, the in-phase and quadrature components of the support-force signal at the chop frequency f_c .
8. **Sense-odd and sense-even components** — $\Delta F_{\text{odd}} = (\Delta F_+ - \Delta F_-)/2$ and $\Delta F_{\text{even}} = (\Delta F_+ + \Delta F_-)/2$, computed from the synchronous observables at the two alignment senses.
9. **Empirical response map (L1)** — a stored function relating drive state and, where relevant, environmental state to a response quantity (response class, force offset, normalized coupling-response coefficient, or observation-model output), together with its uncertainty or covariance, hysteresis and time constants, provenance record, and validity envelope. A map is established by measurement against separate references; it presumes no sign, magnitude, additivity, or scaling and admits zero.
10. **Provenance record** — the identity of the unit or production class, the reference(s) used, date, operator or process, raw-data identifier, fit method, and the validation gates passed.
11. **Validity envelope** — the region of drive and environmental state over which the map was established and validated, including permissible interpolation, together with runtime limits (temperature, orientation, field gradient, vibration, bandwidth, drift rate, waveform distortion) outside which the controller reports out-of-envelope.
12. **Qualified module** — a driven-medium module or production class whose calibration memory holds an empirical response map containing at least one independently validated operating point, its uncertainty, provenance, validity envelope, and safe-off behavior. A module whose map contains only null operating points is qualified as a null reference and may not report L2 quantities.
13. **Response class** — an operational label assigned by the classification map from the signs, magnitudes, and uncertainties of the sense-odd and sense-even components under stated validity criteria (for example: sense-odd positive; sense-odd negative; sense-even dominated; null; unclassified). Response classes are defined by measurement, not by physical account.
14. **Channel partition (λ)** — a hypothesis parameter describing how driven angular momentum may be divided between an intrinsic-alignment ("twist") channel and an orbital-winding ("writhe") channel. It is not defined by the support-force result and is reportable as an estimate λ only under the endpoint-standard condition of Definition 15.
15. **Endpoint standards** — independently established response-class standards for a specimen or operating point in which one channel dominates, recorded in the classification map with

uncertainty. Absent endpoint standards the analyzer reports a response class, not λ .

16. **Absorbed-energy-matched control** — a control state (detuned, phase-scrambled, or sham drive) adjusted so that active and control runs have matched absorbed-energy and temperature histories within a stated tolerance; equality of source power alone is not sufficient.
17. **Reaction frame; load paths** — the mechanically independent structure carrying the alignment-field source, coupling structure, feeds, thermal hardware, and shields, whose support loads are transmitted to a reaction support separate from the force transducer. The body load path (medium → holder → force transducer → base) and the reaction load path (frame → reaction support → base) are both instrumented in qualification.
18. **Normalized coupling-response coefficient (κ)** — for C3, the synchronous observable divided by a stated normalization force (the body's nominal support force $m \cdot g_0$ unless otherwise stated), corrected by the empirical response map, with uncertainty.
19. **Observation model (H)** — for C4, the stored empirical relation $y = H(b_a, s_a, \kappa_local, attitude, T, motion\ state, \dots) + \epsilon$ between the measured synchronous and unmodulated force-channel outputs and the estimator states, where b_a is additive accelerometer bias, s_a is scale factor, and κ_local is the local coupling-response state.
20. **Validity states** — the controller-reported states *valid*, *degraded*, *out-of-envelope*, *invalid* (for C1, *unclassified*), and *safe-off*, as defined in FIG. 10.

THE DRIVEN-MEDIUM MODULE (500) — COMMON PRODUCT ARCHITECTURE (FIG. 1)

Ordered medium (502). The reference body is a polished single-crystal YIG sphere, diameter 2–15 mm (reference 10 mm; volume 0.5236 cm³; mass approximately 2.71 g at 5.17 g/cm³; nominal support force approximately 26.6 mN). Alternatives include other garnets, ferrites, and engineered or composite media of sufficient phase coherence, i.e., media whose population coherence time τ and drive turn rate v satisfy $\tau \cdot v \geq 1$ under the recited alignment field; single-crystal YIG at gigahertz-class drive is expected to exceed this floor by two to three orders of magnitude, to be measured for the specimen and mode selected. The medium is carried in a **specimen holder (503)** of low-loss, nonmagnetic material (quartz, sapphire, PTFE, alumina) that transmits its support force through a stem to the force transducer.

Alignment-field source (504). An independently supported electromagnet, permanent-magnet yoke, or coil-and-yoke assembly producing a static field of commanded magnitude and polarity, reference 1500–2500 Oe for the reference sphere (approximately 0.2 T; e.g., a 76 mm-pole C-frame electromagnet at an approximately 80 mm gap). Shaped pole faces, trim windings, or shim elements provide the field uniformity required by the gradient budget below. Field-sense reversals are performed in separate thermally settled blocks; the field and its first spatial derivatives are remapped at the body and transducer travel volumes after each reversal. In deployed C3/C4 units the source is preferably a permanent-magnet yoke with a trim winding for sense selection by a commanded auxiliary field or by mechanical rotation, as the map for the unit dictates.

Coupling structure and drive chain (506, 508). A coupling structure (TE-mode rectangular cavity, loop-gap resonator, re-entrant resonator, waveguide, or coil) places the body at a mapped antinode of the oscillating drive's magnetic component, with the drive's magnetic axis orthogonal to the alignment field. The reference structure is a rectangular TE₁₀₂ cavity for approximately 5.6 GHz with a below-cutoff stem aperture; the loaded field vector is mapped after installation of body and stem. The drive chain (508) comprises a voltage-controlled source spanning the resonance (reference 4–8 GHz), pulse and waveform modulator, power amplifier (reference 1–5 W for qualification; deployed units are sized to their map), directional coupler, and incident/reflected/transmitted-power detectors. The controller servo-locks to the located precession resonance and records the resonance mode, linewidth, loaded quality factor, and coupling-structure ring-up and ring-down times. Duty factor is controller-limited; higher delivered energy is implemented as bursts separated by recovery intervals.

Force transducer (510). A force sensor (flexure with capacitive or optical readout, load-cell, or balance mechanism) supporting the specimen holder through the defined body load path (511). Its transfer function (gain, phase, bandwidth) is calibrated over the registered force bandwidth against a known force injection; its electronics are placed outside the high-field region or coupled through a nonmagnetic force-transfer member. For the reference body a 0.01 mg-force-equivalent threshold is approximately 98 nN.

Reaction frame and load paths (512, 513). The alignment-field source, coupling structure, probe, feed lines, thermal hardware, and shields are carried by a reaction frame mechanically independent of the force transducer, and their support loads pass through a **reaction load path (513)** to a reaction support and thence to the common base. A reaction-force sensor under the frame records that channel. Where the complete head is placed on an outer platform sensor, the calibrated sum of all instrumented support paths supplies the external account. The disclosure makes no claim that an enclosure can give itself a sustained net external force by internal rearrangement: the body load path and the reaction load path are both measured, and any hypothesized external reaction (e.g., against the local coupling environment) is a L2 interpretation reported only under the validity gates.

Synchronized acquisition (514). A common clock timestamps: support force; reaction force; incident, reflected, and transmitted drive power; cavity, body, stem, coil, transducer, and ambient temperature; pressure; three-axis static field and mapped force-axis gradient; vibration and three-axis acceleration; strain at load-path and feed supports; orientation or attitude; position and time registration (C3/C4); and the commanded drive-state sequence. In-phase, quadrature, and harmonic components at the toggle or chop frequency are recorded.

Controller (516) and calibration memory (518). The controller commands drive state within the validity envelope, executes the product's operating sequence (toggle, reversal, chop, waveform, control states), monitors runtime witnesses, applies the stored empirical response map, propagates uncertainty, and reports the validity state (FIG. 10). The calibration memory holds the map, provenance record, validity envelope, hysteresis and thermal coefficients, zero-check history, and recalibration schedule.

Environmental sensors (520). Temperature (multiple points), pressure, three-axis field and gradient witness, accelerometer, tilt/attitude, acoustic/vibration, and drive-power monitors supply the runtime witnesses tested against the validity envelope.

Magnetic-gradient budget (all products). For a saturated reference sphere with moment approximately $0.074 \text{ A} \cdot \text{m}^2$, a force-axis gradient of approximately $1.32 \text{ } \mu\text{T/m}$ imitates 98 nN . Because a gradient force can be resonant, synchronous, and reversal-sensitive, every product maps or dithers the force-axis component of $\nabla(\mu \cdot \mathbf{B})$ at the body and either extrapolates the response to zero gradient or bounds the residual below a preregistered fraction (preferably one third) of the product's reporting threshold; otherwise the state is out-of-envelope.

LAYERED SCOPE (FIG. 2)

- **L0 — raw synchronous observable.** ΔF_s , ΔF_{odd} , ΔF_{even} (toggle) or the in-phase and quadrature components at f_c (chop), each with statistical uncertainty from the recorded cycles and drift model. L0 is always reportable.
- **L1 — calibrated response coefficient.** The map-corrected quantity: for C1 the normalized odd/even response vector; for C2 the force offset in newtons; for C3 κ ; for C4 the observation-model output. Reportable when the drive and environmental state lie inside the validity envelope.
- **L2 — inferred quantity.** Response class or $\hat{\chi}(C1)$; certified source output for transfer to a sensor under calibration (C2); coupling or gravimetric interpretation, gradient, or source parameters (C3); accelerometer bias separated from scale factor and local coupling response (C4). Reportable only when the validity gates pass and the stated calibration or model conditions are met; otherwise the controller reports unclassified/invalid.
- **L3 — operational use.** Material or operating-point selection (C1); calibration of a separate force sensor (C2); survey map generation and conditional inversion (C3); corrected inertial navigation within a bounded-error envelope (C4).

QUALIFICATION, CALIBRATION, AND VALIDATION (FIG. 3)

Factory qualification vs. deployed operation. Qualification is performed once per unit or production class with the full control matrix; deployed operation uses the stored map and a reduced set of runtime witnesses and self-tests. A deployed unit never reports a L2 quantity its map does not support.

Empirical-map procedure. (a) *Acquire*: execute a randomized sequence of drive states and control states across the intended envelope against the relevant separate reference (C1: reference specimens and endpoint standards where available; C2: a separate force reference; C3: calibration sites or known reference configurations; C4: a qualification turntable and reference accelerometer), recording all synchronized channels. (b) *Fit*: fit the response as a function of drive and environmental state with a preregistered model class (lookup with interpolation, polynomial, or physics-agnostic regression), estimating covariance, hysteresis, and time constants. (c) *Validate*: test the fitted map on held-out operating points and on a separately assembled replication where available; apply the validity gates

below; record residuals. (d) *Store*: write map, covariance, provenance, validity envelope, and safe-off parameters to the calibration memory. (e) *Update*: runtime zero checks and reference checks are logged; drift beyond tolerance shrinks the envelope or triggers recalibration. (f) *Apply*: the controller evaluates the map at the current state, propagates uncertainty, and reports the layer permitted by the current validity state.

Control matrix (qualification; subset at runtime). Randomized ON/OFF ordering; detuned drive at matched absorbed energy; phase-scrambled or sham drive; non-alignable and magnetic reference specimens of matched geometry, mass, thermal, and mechanical response; independent reversal of alignment sense and of body orientation; mapped field gradient and contemporaneous magnetization; body, mount, transducer, and ambient temperature monitoring; vacuum or quantified buoyancy and convection correction; accelerometer and acoustic monitoring; cable-routing and drive-termination swaps; force-transducer transfer-function and phase calibration; in-phase, quadrature, and harmonic reporting at two or more modulation frequencies; positive-control force injections of known magnitude and phase; drift and Allan deviation analysis with preregistered thresholds.

Validity gates (V1–V8). V1 repeatability across blocks and days; V2 resonance contrast against the absorbed-energy-matched detuned control; V3 material contrast against reference specimens and swaps; V4 magnetic-force exclusion by the gradient budget; V5 thermal, radiometric, buoyancy, and convection exclusion; V6 mechanical and acoustic exclusion, including feed strain and acceleration-coupled force; V7 statistical validity after multiple-comparison control with in-phase/quadrature/harmonic consistency; V8 replication on a second unit or laboratory for any map used to report a L2 quantity to a third party.

Environmental and dynamic envelopes. The map records and the controller enforces: temperature range and thermal coefficients of body, mount, and transducer; permissible orientation and tilt; force-axis field-gradient limit; vibration and acceleration spectral limits; force-channel bandwidth and phase response; drift rate and zero-check interval; hysteresis of the response with drive history; and waveform distortion (harmonic content and slew) for modulated operation. Out-of-envelope conditions cause the controller to withhold L2 reporting and to flag the record.

Cryogenic embodiments. Where a product operates the medium at or below 77 K, the cryostat is on the reaction frame; the body load path passes through a thermally anchored, low-conductance stem; boil-off and pressure fluctuation, cable and thermal-strap forces, thermal-gradient drift, and settling time after fill are witnessed and gated.

DETAILED DESCRIPTION — C1: CHANNEL-RESPONSE ANALYZER (FIGS. 4–5)

Product. A bench analyzer (550) packaging the module of FIG. 1 with a specimen-exchange holder (503), evacuable enclosure, alignment-field source with commanded sense reversal, resonance-locked drive, force transducer, reaction frame, synchronized acquisition, controller (516), calibration memory (518) holding a classification map (552), and a report output (556). Specimens are the ordered media to be characterized—crystals, ferrites, films on carriers, composite or engineered media—at selectable temperature and alignment magnitude.

Operating sequence. Establish alignment at a commanded magnitude (preferably measured saturation); locate and lock the precession resonance; toggle the drive ON/OFF in a randomized sequence at a period long against transducer settling while holding alignment sense fixed; record the synchronized channels; repeat in a separate settled block at the opposite sense; interleave absorbed-energy-matched detuned control, sham drive, and reference-specimen blocks; compute ΔF_+ and ΔF_- and decompose into ΔF_{odd} and ΔF_{even} with uncertainties; test V1–V7; evaluate the classification map; report.

Classification and what is reported. The classification map (552) assigns a response class from the signed, normalized (ΔF_{odd} , ΔF_{even}) vector and its uncertainty relative to the validity criteria. The map contains a **null region** (response within uncertainty of zero), an **unclassified region** (response outside null but validity criteria unmet or the vector outside mapped classes), and mapped classes. The applicant's hypothesis (FIG. 5) is that an intrinsic-alignment-dominated route and an orbital-winding-dominated route produce sense-odd responses of opposite sign; the analyzer *tests* that hypothesis on each specimen and reports the class it measures. **λ condition.** Only when the map holds endpoint standards (Definition 15)—independently established response-class standards for each dominated route—does the processor interpolate a channel-partition estimate $\hat{\lambda}$ with uncertainty; absent such standards it reports the response class. If the response is modeled as $\Delta F = J[a_w + (a_t - a_w)\lambda]$ with unknown channel coefficients a_t , a_w and driven angular momentum J , a quantitative λ requires the endpoint standards to fix a_t and a_w for the operating point; this is why λ is conditional.

Spectroscopy and operating-point selection (L3). The controller sweeps alignment magnitude, pulse width, drive power, duty, or temperature and records the **response spectrum** (L1) of the specimen; where endpoint standards exist it also reports a **calibrated partition spectrum** (L2). Comparing spectra across compositions, temperatures, and bias conditions selects media and operating points for driven-medium modules used in the C2–C4 products and in companion applications. Variants: swept-bias spectroscopy; pulse-width dependence; cryogenic narrowing at or below 77 K with the cryogenic provisions above.

DETAILED DESCRIPTION — C2: PROGRAMMABLE MICROFORCE SOURCE AND FORCE-SENSOR CALIBRATOR (FIG. 6)

Product. A programmable force source (570) comprising the module of FIG. 1 in a sealed head with a **mounting interface (571)** through which the body's support force is transmitted to the force sensor under calibration (576); a reaction frame whose loads pass to a separate reaction support (513) and not through the mounting interface; an independent **source-monitor force channel (572)** reading the body's support force in parallel with, and independently of, the sensor under calibration; a controller (516) commanding sign (by alignment sense or drive state), amplitude, waveform, and modulation frequency; a **calibration memory (518)** holding the empirical transfer map $F_{\text{out}} = \mathcal{C}(u, f, B, T, \text{orientation, history, } \dots)$ with covariance, provenance, validity envelope, hysteresis, thermal coefficients, and zero-check history; and an output interface (578) reporting the commanded force-offset value and its uncertainty to the data system of the sensor under calibration.

Reaction path. In the calibrator configuration the body load path terminates on the sensor under calibration; the reaction load path terminates on the reaction support. Both are instrumented in qualification so that the offset reported at the mounting interface is the measured body-load-path quantity and is not confounded with frame reaction.

Calibration against a separate reference; no circularity. The transfer map is established by mounting the module on a **separate force reference (574)** of stated uncertainty (deadweight comparator, electrostatic force balance, or Kibble-type reference at the relevant force class), commanding the drive-state set across the intended envelope, and fitting F_{out} against the reference reading with the source monitor recorded simultaneously. The sensor to be calibrated never serves as the reference for its own calibrator. Thereafter the source monitor witnesses each commanded output; deviation of the monitor from the map beyond tolerance invalidates the transfer and flags recalibration.

Traceability chain (corrected). Drive parameters are frequency-anchored: the volt is realized via the Josephson constant $K_J = 2e/h = 483\,597.848\text{ GHz/V}$, and $e/h = 241.798\,924\text{ THz/V}$ (a turn rate *per volt*, becoming a turn rate only when multiplied by an actual voltage); the kilogram is realized via the Kibble balance from electrically measured watts. Electrical traceability of the drive does **not** by itself make the emitted force traceable. The chain disclosed here is: drive parameters (frequency, power, duty, waveform, field) → empirical transfer map established against a separate SI-traceable force reference → commanded force offset with propagated uncertainty → sensor under calibration. The force reference, not the drive electronics, anchors the newton.

Commanded operation. Within the validity envelope the controller commands sign-reversible offsets, steps, square or sinusoidal modulation at DC to kilohertz class, and selected or user-defined waveforms **within the characterized amplitude, bandwidth, slew-rate, and distortion envelope**; commands outside the envelope are inhibited or flagged. Periodic zero checks (drive off; sense reversed) and reference checks are interleaved. Uses: gain and linearity mapping of balances, load cells, and microforce sensors; dynamic (frequency-response and phase) characterization by stepped modulation frequency; remote, sign-reversible checking of installed sensors without exchanging deadweights.

DETAILED DESCRIPTION — C3: ACTIVE COUPLING SURVEY AND GRADIOMETRY SYSTEM (FIGS. 7–8)

Product. A survey unit (590) comprising one or more driven reference units, each the module of FIG. 1 in a ruggedized head; a **modulator (592)** chopping the drive at a selected frequency f_c under a **common clock (593)**; a **synchronous detector (594)** extracting in-phase and quadrature components at f_c and at selected harmonics; **position and time registration (596)** (satellite, odometry, inertial, depth encoder, or fixed-station timing); environmental sensors (520) including attitude, acceleration, speed, three-axis field, temperature, pressure; a calibration memory (518) holding the response map relating the synchronous observable to the normalized coupling-response coefficient κ (Definition 18) with uncertainty, provenance, and validity envelope; and a **survey processor (598)** producing the coefficient map (L1/L2) with uncertainty and validity flags.

What is measured and what is inferred. Synchronous detection rejects off-frequency noise; it does not establish causation, and thermal, mechanical, inertial, or field effects coherent with f_c pass through it. The unit therefore reports layered outputs: (L0) the synchronous response at f_c ; (L1) $\kappa = \Delta F_{\text{synchronous}} / F_{\text{normalization}}$ corrected by the map; (L2) a **coupling or gravimetric interpretation**—variation of a local coupling coefficient (the quantity the extant art expresses through G , $\text{m}^3 \text{kg}^{-1} \text{s}^{-2}$, and reads as g , m s^{-2}), density anomaly, or source parameters—only when the map is established at calibration sites or against known source configurations, the model assumptions are stated, and corroborating passive-gravimeter or density information is available where the inversion requires it. A single force reading does not separate G , source mass, and geometry; the product does not purport to do so.

Survey method. Calibrate the unit(s) at a calibration site or against a known reference configuration (mapping κ against the reference over the environmental envelope); chop at f_c with drive-frequency interleaving; perform stationary controls before and after each traverse; traverse with position/time registration while recording synchronized channels and environmental witnesses; execute drive-off and detuned control intervals; repeat reciprocal traverses; compute κ with uncertainty at each registered position/time; test the validity gates; generate the survey map with validity flags; optionally run blinded source/no-source trials at test masses.

Gradiometer (FIG. 8). Two or more units on fixed, measured baselines L with common-clock, phase-coherent modulation and matched gain/phase calibration. Two units estimate one baseline-projected gradient, $\Gamma_L \approx (R_B - R_A)/L$, and reject common-mode platform response. Localization of an unrestricted three-dimensional source from one simultaneous difference is not possible; the product reports localization only when supplied with multiple baselines, three or more spatial channels, traverse geometry, or explicit stated assumptions about source depth, shape, or direction, and reports the associated uncertainty and the assumptions used.

Embodiments. *Vehicle survey unit (600):* ground, marine, or airborne carrier; attitude, acceleration, vibration, and speed recorded; inertial compensation of the force channel over the survey bandwidth; reciprocal traverses. *Borehole sonde (602):* pressure housing; temperature, pressure, cable-tension, and sonde-orientation sensors; wall-coupling standoff or clamp; downhole power and telemetry; vibration isolation; depth registration. *Stationary observatory (604):* continuous time series with drift and Allan-deviation reporting; comparison with passive gravimeter and tidal models as corroborating channels; secular monitoring.

DETAILED DESCRIPTION — C4: COUPLING-REFERENCED INERTIAL-NAVIGATION AND BIAS-CALIBRATION SYSTEM (FIG. 9)

Product. A navigation system (620) comprising an **inertial measurement unit (622)** with accelerometers and gyroscopes; a **driven reference module (624)** (the module of FIG. 1) co-mounted in a known geometric relation; modulator and synchronous detector; attitude, temperature, and motion-state inputs; a memory (518) holding the **observation model H** (Definition 19) with provenance and validity

envelope; an **estimator (626)** for accelerometer bias, scale factor, and local coupling-response state; a **navigation processor (628)** producing corrected inertial state with propagated uncertainty; and **state logic (630)** reporting valid, degraded, or invalid navigation with error bounds.

Observation model. The un-modulated support force of the reference body, divided by its mass, is itself a specific-force reading along the transducer axis with its own bias b_F and scale s_F . The commanded, map-calibrated force step ΔF_{cmd} (from the C2-type transfer map) provides an **in-situ absolute scale reference** for the force channel and, through the known geometry, for the co-registered accelerometer axis. The model is $y = [y_{\text{sync}}, y_{\text{unmod}}, y_{\text{IMU}}] = H(b_a, s_a, b_F, s_F, \kappa_{\text{local}}, \text{attitude}, T, \text{motion state}) + \varepsilon$, learned at qualification on a turntable against a reference accelerometer and refined in the field.

Observability and separation. The disclosure does not assert that the chopped channel alone determines additive accelerometer bias. Separation is achieved by: (i) *scale* — the commanded step calibrates s_F and, by co-registration during the same interval, s_a ; (ii) *bias* — orientation reversal of the module or of the IMU-module cluster about a horizontal axis reverses the sign of the specific-force term while additive biases b_a and b_F do not reverse, so their difference and, with the module's drive-off/sense-reversed zero check bounding b_F , b_a itself become observable during quasi-static intervals; (iii) *local coupling response* — κ_{local} is estimated as a separate state only when the observation model marks it separable from b_a and s_a under the recorded motion state and the validity gates pass. The estimator performs an **observability (rank) test** on H linearized about the current state and motion history before updating a state; unobservable states are held with growing covariance.

Motion-state discrimination. Gyroscope and accelerometer statistics classify intervals as stationary, quasi-static (specific force within tolerance of the modeled local value with bounded rotation), or dynamic. Bias updates are performed only in stationary or quasi-static intervals; scale updates from commanded steps may proceed in any interval within the force channel's dynamic envelope.

Initialization, bounded error, and degraded logic. Position, velocity, and attitude are initialized from an external fix or stationary leveling and gyrocompassing; the system propagates covariance and declares *valid* while predicted error remains within the characterized envelope, *degraded* when a required update (quasi-static interval, zero check, or reversal) has not occurred within its characterized interval or an environmental witness is out of envelope, and *invalid* when the error bound is exceeded or the module reports out-of-envelope. Operation without contemporaneous satellite navigation input (GNSS: the GPS, GLONASS, Galileo, and BeiDou class) is a conditional mode for a characterized interval and error envelope; the disclosure does not assert unbounded GNSS-free navigation, and gyroscope drift and heading error are not corrected by this channel unless separately aided.

PROPHETIC EXAMPLES (prospective; every number follows from the stated mass, noise model, bandwidth, and integration time; no result is represented as achieved)

P1 — Analyzer on a 10 mm YIG specimen. A 10 mm sphere (mass ≈ 2.71 g; support force ≈ 26.6 mN) at 2000 Oe would be toggled at a period long against a transducer whose one-cycle 1σ noise is 0.01 mg-force-equivalent (≈ 98 nN). Under an independent white-noise model with drift removed by ON/OFF

differencing, 100 toggle cycles would give $\sigma \approx 9.8$ nN and a 3σ declaration threshold ≈ 29 nN (≈ 0.003 mg-force; $\approx 1.1 \times 10^{-6}$ of the support force). If instead 0.01 mg-force is the final 100-cycle resolution, the 3σ threshold is 0.03 mg-force (≈ 294 nN). Correlated noise or drift would raise either threshold; the analyzer's Allan-deviation report would state the realized value. The gradient budget requires the force-axis residual gradient below ≈ 0.44 $\mu\text{T/m}$ for the 98 nN class. The expected report is a response class with uncertainty; χ would be reported only if endpoint standards had been established.

P2 — Programmable force source. A module would be commanded to ± 1 mg-force (± 9.80665 μN ; $\approx 3.7 \times 10^{-4}$ of the reference body's 26.6 mN support force) square-wave steps at 10 Hz for balance linearity mapping. With a source-monitor floor of 98 nN/ $\sqrt{\text{Hz}}$ and lock-in integration of 100 s (equivalent noise bandwidth $1/(2 \cdot 100 \text{ s}) = 5$ mHz), the monitored step amplitude would have $\sigma \approx 6.9$ nN ($\approx 0.07\%$); the certified uncertainty would be the combination of that statistic, the separate force reference's stated uncertainty, and the map's covariance at the commanded state. Source-monitor and sensor bandwidth would be ≥ 100 Hz for 10 Hz steps. Whether a given module produces a nonzero mapped offset is established by its qualification, not assumed.

P3 — Survey unit. A unit with the 2.71 g reference body (normalization force 26.6 mN) chopped at 1 kHz would require a force channel with bandwidth ≥ 2 kHz. A design target of 10^{-9} relative amplitude spectral density corresponds to ≈ 26.6 pN/ $\sqrt{\text{Hz}}$ on the support-force channel; a conservative target of 10^{-7} corresponds to ≈ 2.66 nN/ $\sqrt{\text{Hz}}$. At 10 s integration per sample (equivalent noise bandwidth 50 mHz) the 3σ minimum detectable change in κ would be $\approx 6.7 \times 10^{-10}$ (≈ 18 pN) for the design target and $\approx 6.7 \times 10^{-8}$ (≈ 1.8 nN) for the conservative target—the former of order 1 μGal -class and the latter of order 0.1 mGal-class relative to go, for comparison with extant gravimetric practice only. A 10 km traverse at 10 m/s would yield 100 samples at 100 m spacing in 1000 s. The realized sensitivity is set by the selected transducer's measured noise floor at f_c and by the vehicle's inertial residual after compensation; the gravimetric interpretation of any mapped variation is conditional on the calibration-site map.

P4 — Navigation system. An accelerometer with 100 μg additive bias ($\approx 9.81 \times 10^{-4}$ m/s²) uncorrected produces position error $\frac{1}{2}bt^2 \approx 6.4$ km after one hour along that axis. If the reversal-and-zero-check procedure and estimator reduce the residual bias to 10 μg , the accelerometer-bias contribution falls to ≈ 640 m per hour. A 10 μg bias on the 2.71 g reference body is ≈ 266 nN; a 98 nN/ $\sqrt{\text{Hz}}$ force channel resolves it (1σ) in ≈ 0.14 s, so the limit is set by the force channel's own zero stability and the reversal cadence, not by its noise. Heading and gyroscope-drift contributions are unchanged by this channel. The system would report *valid* while its propagated bound remains within the characterized envelope and *degraded* thereafter.

POAMS-NATIVE OUTPUT LANE

Calibration maps and response classes provide conservative operational outputs, but they do not replace the native quantity under test. The analyzer may report the POAMS twist/writhe partition, the force source may report a constraint-bill change, and the survey instrument may report a local coupling-ledger

change under the stated POAMS definitions. Conventional reaction-path, gravimeter, and Newtonian quantities remain tagged controls or comparison species and are not ontological prerequisites.

CLAIMS

1. A channel-response analyzer for characterizing a specimen comprising an ordered medium, the analyzer comprising: a specimen holder; a force transducer supporting the specimen holder through a defined body load path and producing a support-force signal; an alignment-field source configured to apply to the specimen a static alignment field of commanded magnitude and sense; a coupling structure and drive source configured to apply to the specimen an oscillating drive at a located precession resonance of the specimen's aligned internal-circulation population; a reaction frame carrying the alignment-field source and the coupling structure and transmitting their support loads through a reaction load path separate from the body load path; a synchronized acquisition unit recording the support-force signal together with drive state, absorbed drive power, alignment-field magnitude and sense, force-axis field gradient, specimen temperature, and vibration; a controller configured to toggle the oscillating drive between on and off states while holding the alignment sense fixed, to reverse the alignment sense in a separate settled block, and to execute at least one absorbed-energy-matched control state; a processor configured to compute from the recorded support-force signal a synchronous differential force for each alignment sense and to decompose the synchronous differential forces into a sense-odd component and a sense-even component; a memory storing a classification map relating the sense-odd component, the sense-even component, and the recorded drive state to a set of response classes, the classification map having a recorded provenance and a validity envelope; and an output configured to report a response class with an uncertainty when the recorded drive state lies within the validity envelope and predetermined validity criteria are satisfied, and otherwise to report an unclassified state.
2. The analyzer of claim 1, wherein the classification map further stores endpoint standards comprising independently established response-class standards for an intrinsic-alignment-dominated route and for an orbital-winding-dominated route, and the processor is configured to report a channel-partition estimate with uncertainty only when the endpoint standards are present and the predetermined validity criteria are satisfied.
3. The analyzer of claim 1, wherein the controller is configured to repeat the toggling and the reversing at a plurality of alignment-field magnitudes, drive pulse widths, drive powers, or specimen temperatures, and the output reports a response spectrum of the specimen comprising the sense-odd and sense-even components as a function of the varied quantity.
4. The analyzer of claim 1, wherein the predetermined validity criteria comprise: resonance contrast against the absorbed-energy-matched control state; material contrast against at least one non-alignable reference specimen and at least one magnetic reference specimen; exclusion of a force

imitated by the recorded force-axis field gradient acting on a measured magnetic moment of the specimen; thermal exclusion; repeatability across blocks; and consistency of in-phase, quadrature, and harmonic components at two or more toggle frequencies.

5. The analyzer of claim 1, further comprising a cryostat carried by the reaction frame and configured to hold the specimen at or below 77 K, the body load path passing through a thermally anchored stem, and the synchronized acquisition unit further recording cryogen boil-off pressure, stem and strap forces, and thermal-gradient drift as validity witnesses.
6. The analyzer of claim 1, packaged as a bench instrument comprising an evacuable enclosure, a specimen-exchange interface, a reaction-force sensor under the reaction frame, and a report output that records the response class, its uncertainty, the drive state, the validity state, and the provenance of the classification map for each specimen.
7. A method of operating a channel-response analyzer to characterize a specimen comprising an ordered medium, the method comprising: mounting the specimen on a force transducer through a defined body load path while an alignment-field source and a coupling structure are carried by a reaction frame having a reaction load path separate from the body load path; applying a static alignment field of a first sense and measuring an alignment state of the specimen; locating a precession resonance of the specimen's aligned internal-circulation population and locking an oscillating drive to it; toggling the oscillating drive between on and off states in a randomized sequence at the first sense while recording a support-force signal, drive state, absorbed drive power, force-axis field gradient, specimen temperature, and vibration; repeating the toggling in a separate settled block at a second, opposite alignment sense; executing at least one absorbed-energy-matched control state and at least one reference-specimen state; computing a synchronous differential force for each alignment sense and decomposing the synchronous differential forces into a sense-odd component and a sense-even component with uncertainties; testing predetermined validity criteria; evaluating a stored classification map having a recorded provenance and a validity envelope at the recorded drive state; and reporting a response class with uncertainty when the validity criteria are satisfied and the drive state lies within the validity envelope, and otherwise reporting an unclassified state.
8. The method of claim 7, further comprising repeating the method for a plurality of specimens, compositions, temperatures, or alignment-field magnitudes, comparing the reported response classes or the sense-odd and sense-even components, and selecting a medium or an operating point for a driven-medium module based on the comparing.
9. The method of claim 7, wherein the stored classification map contains endpoint standards for an intrinsic-alignment-dominated route and an orbital-winding-dominated route, and the reporting comprises reporting a channel-partition estimate with uncertainty only when the endpoint standards are present and the validity criteria are satisfied.

10. A programmable force source for calibrating a force sensor, comprising: a driven module comprising a body of ordered medium, an alignment-field source configured to apply to the body a static alignment field of commanded magnitude and sense, and a coupling structure and drive source configured to apply to the body an oscillating drive at a located precession resonance of the body's aligned internal-circulation population; a mounting interface through which a support force of the body is transmitted to a force sensor under calibration; a reaction frame carrying the alignment-field source and the coupling structure and transmitting their support loads to a reaction support separate from the mounting interface; a source-monitor force channel reading the support force of the body independently of the force sensor under calibration; a calibration memory storing an empirical transfer map relating a commanded drive state, including a sign selection, an amplitude, a waveform, and a modulation frequency, to an output force offset, the transfer map having been determined by measurement against a separate force reference and stored with an uncertainty, a provenance record, and a validity envelope; a controller configured to command the sign selection, amplitude, waveform, and modulation frequency within the validity envelope, to compute from the transfer map a commanded force-offset value with uncertainty, and to inhibit or flag a command outside the validity envelope; and an output interface configured to report the commanded force-offset value and its uncertainty to a data system associated with the force sensor under calibration.
11. The programmable force source of claim 10, wherein the waveform is selected from steps, square waves, sinusoids, and user-defined waveforms within a characterized amplitude, bandwidth, slew-rate, and distortion envelope stored in the calibration memory.
12. The programmable force source of claim 10, wherein the controller is configured to interleave zero checks, comprising a drive-off state and a sense-reversed state, and reference checks with commanded operation, to compare the source-monitor force channel with the transfer map, and to invalidate the commanded force-offset value and flag recalibration when the comparison exceeds a stored tolerance.
13. The programmable force source of claim 10, wherein the calibration memory further stores hysteresis with drive history, thermal coefficients, and orientation dependence of the output force offset, and the controller is configured to compensate the commanded force-offset value therefor or to restrict the validity envelope accordingly.
14. A method of calibrating a force sensor with a programmable force source comprising a driven module having a body of ordered medium, an alignment-field source, a coupling structure and drive source, a reaction frame, and a source-monitor force channel, the method comprising: mounting the driven module on a separate force reference of stated uncertainty; commanding a set of drive states, each comprising a sign selection, an amplitude, a waveform, and a modulation frequency, while recording an output force offset read by the separate force reference and by the source-monitor force channel; fitting an empirical transfer map relating the commanded drive states to the output force offset with uncertainty; storing the transfer map with a provenance record

and a validity envelope in a calibration memory; mounting the driven module on the force sensor under calibration such that a support force of the body is transmitted to the force sensor and reaction loads of the alignment-field source and the coupling structure are transmitted to a separate reaction support; commanding sign-reversible modulated force offsets within the validity envelope while recording an output of the force sensor synchronized with the commanded drive states and with the source-monitor force channel; comparing the recorded output of the force sensor with commanded force-offset values computed from the transfer map; reporting a calibration of the force sensor comprising at least one of gain, linearity, and frequency response with propagated uncertainty; and reporting the calibration as invalid where the source-monitor force channel deviates from the transfer map beyond a stored tolerance.

15. The method of claim 14, wherein the commanding of sign-reversible modulated force offsets comprises stepping the modulation frequency through a plurality of values within the validity envelope, and the reported calibration comprises an amplitude and phase frequency response of the force sensor.
16. The method of claim 14, further comprising interleaving drive-off and sense-reversed zero checks with the commanded force offsets and correcting the reported calibration for a drift determined from the zero checks.
17. An active coupling survey system comprising: at least one driven reference unit comprising a reference body of ordered medium, an alignment-field source configured to apply to the reference body a static alignment field, a coupling structure and drive source configured to apply to the reference body an oscillating drive at a located precession resonance of the reference body's aligned internal-circulation population, a force transducer supporting the reference body through a defined body load path, and a reaction frame carrying the alignment-field source and the coupling structure through a separate reaction load path; a modulator configured to chop the oscillating drive at a selected chop frequency under a common clock; a synchronous detector configured to extract in-phase and quadrature components of the force transducer's signal at the chop frequency; a position and time registration input; environmental sensors comprising at least temperature, three-axis field, acceleration, and attitude sensors; a calibration memory storing an empirical response map relating the extracted components and outputs of the environmental sensors to a normalized coupling-response coefficient with uncertainty, the response map having a provenance record and a validity envelope established at a calibration site or against a known reference configuration; and a survey processor configured to compute the normalized coupling-response coefficient with uncertainty at each registered position or time, to test predetermined validity criteria, and to produce a survey output comprising the coefficient, its uncertainty, and a validity flag at each registered position or time.
18. The survey system of claim 17, wherein the survey processor is further configured to report a coupling-coefficient, gravimetric, density, or source interpretation of the coefficient only when the response map was established against a known reference configuration for that interpretation,

stated model assumptions are recorded with the output, and any required corroborating passive-gravimeter or density input is present.

19. The survey system of claim 17, comprising two or more driven reference units mounted on fixed measured baselines and chopped phase-coherently under the common clock, the survey processor being configured to apply matched gain and phase calibration to the units, to compute a baseline-projected gradient of the coefficient from each pair, and to report a source localization only when multiple baselines, three or more spatial channels, traverse geometry, or stated source assumptions are provided, together with the assumptions used and the resulting uncertainty.
20. The survey system of claim 17, packaged as a vehicle survey unit further comprising a speed input and a vibration sensor, the survey processor being configured to compensate the force transducer's signal for recorded acceleration and attitude over a survey bandwidth and to record reciprocal traverses and stationary control intervals before and after each traverse.
21. The survey system of claim 17, packaged as a borehole sonde comprising a pressure housing, a wall-coupling standoff or clamp, sensors for pressure, cable tension, and sonde orientation, a downhole power supply, a telemetry link, and a depth registration input.
22. The survey system of claim 17, packaged as a stationary observatory, the survey processor being configured to produce a continuous time series of the coefficient with drift and Allan-deviation reporting and to record a corroborating passive gravimeter or tidal-model channel.
23. A method of surveying with an active coupling survey system having at least one driven reference unit comprising a reference body of ordered medium, an alignment-field source, a coupling structure and drive source, a force transducer supporting the reference body, and a reaction frame, the method comprising: calibrating the at least one driven reference unit at a calibration site or against a known reference configuration to establish an empirical response map relating a synchronous force response to a normalized coupling-response coefficient with uncertainty, and storing the response map with a provenance record and a validity envelope; applying an oscillating drive to the reference body at a located precession resonance and chopping the oscillating drive at a selected chop frequency under a common clock; recording stationary control intervals before a traverse or observation period; recording, along the traverse or over the observation period, in-phase and quadrature components of the force transducer's signal at the chop frequency together with position and time registration and environmental sensor outputs; executing drive-off and detuned control intervals; recording stationary control intervals after the traverse or observation period; computing the normalized coupling-response coefficient with uncertainty at each registered position or time from the response map; testing predetermined validity criteria; and generating a survey output comprising the coefficient, its uncertainty, and a validity flag at each registered position or time.
24. The method of claim 23, further comprising interleaving two or more drive frequencies, repeating the traverse in a reciprocal direction, and executing blinded source-present and source-absent trials at a known test mass.

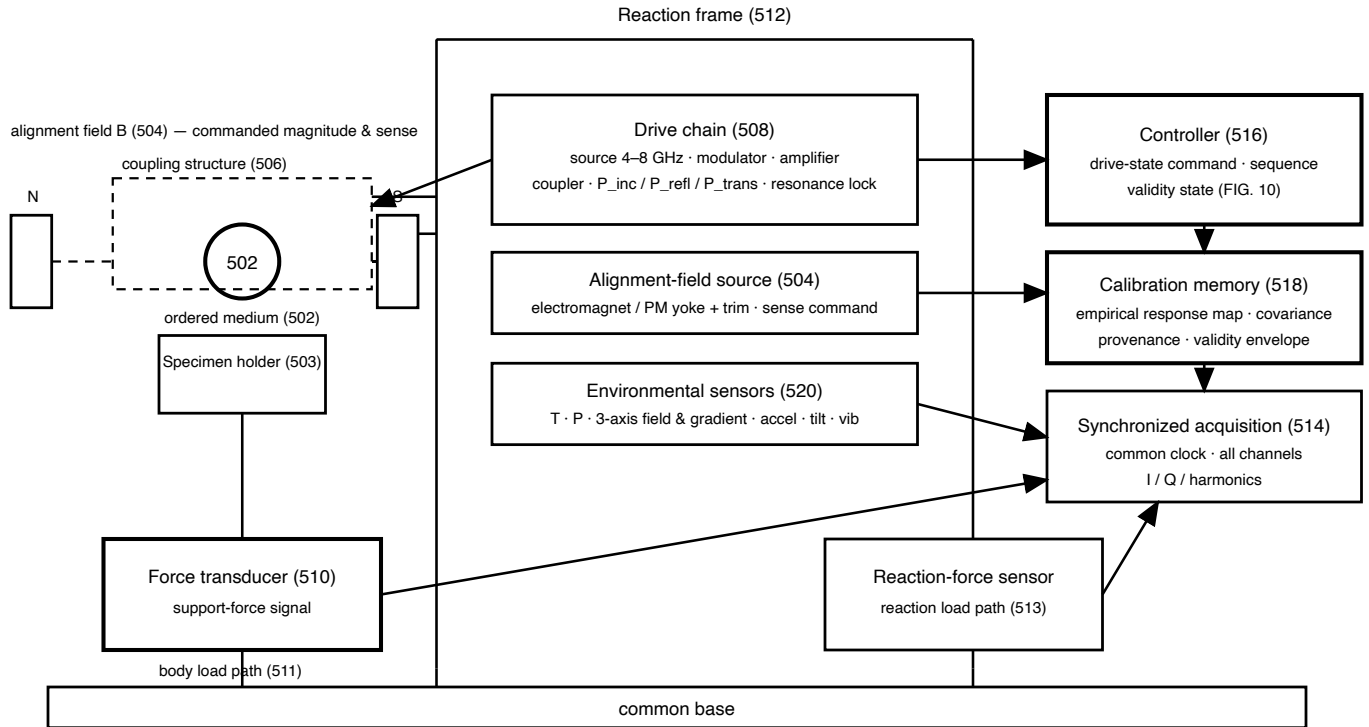
25. The method of claim 23, further comprising recording the in-phase and quadrature components from two or more driven reference units on measured baselines or from repeated positions along the traverse, and inverting the recorded components for a source parameter only when the measured baselines, the repeated positions, or a stated source assumption render the inversion determinate, and reporting the assumption and the uncertainty with the source parameter.
26. A coupling-referenced inertial-navigation system comprising: an inertial measurement unit comprising accelerometers and gyroscopes; a driven reference module co-mounted with the inertial measurement unit in a known geometric relation and comprising a reference body of ordered medium, an alignment-field source configured to apply to the reference body a static alignment field of commanded sense, a coupling structure and drive source configured to apply to the reference body an oscillating drive at a located precession resonance of the reference body's aligned internal-circulation population, a force transducer supporting the reference body through a defined body load path, and a reaction frame carrying the alignment-field source and the coupling structure through a separate reaction load path; a modulator configured to chop or step the oscillating drive; a synchronous detector configured to extract a synchronous component of the force transducer's signal and to output an unmodulated component of the force transducer's signal; attitude, temperature, and motion-state inputs; a memory storing an empirical observation model relating the synchronous component, the unmodulated component, and an output of the inertial measurement unit to an additive accelerometer bias, an accelerometer scale factor, a force-channel bias, a local coupling-response state, attitude, temperature, and motion state, the observation model having a provenance record and a validity envelope; an estimator configured to perform an observability test on the observation model under the motion-state input and to estimate the additive accelerometer bias separately from the accelerometer scale factor and from the local coupling-response state using a drive-off zero-check state and an orientation-reversal state of the driven reference module; a navigation processor configured to produce a corrected inertial state with propagated uncertainty; and state logic configured to report a valid state while the propagated uncertainty remains within a characterized error envelope, a degraded state when a required update has not occurred within a characterized interval or an environmental input lies outside the validity envelope, and an invalid state when the error envelope is exceeded, the system being operable without contemporaneous satellite-navigation input for the characterized interval and error envelope.
27. The navigation system of claim 26, wherein the memory further stores an empirical transfer map of the driven reference module determined against a separate force reference, and the estimator is configured to use a commanded force step computed from the transfer map as an in-situ scale reference for the force transducer and, through the known geometric relation, for a co-registered axis of the accelerometers.
28. The navigation system of claim 26, wherein the motion-state input classifies intervals as stationary, quasi-static, or dynamic from gyroscope and accelerometer statistics, and the estimator updates the additive accelerometer bias only during stationary or quasi-static intervals.

29. The navigation system of claim 26, wherein the estimator is configured to hold the local coupling-response state with growing covariance and to withhold reporting it whenever the observability test does not mark it separable from the additive accelerometer bias and the accelerometer scale factor.
30. A method of operating a coupling-referenced inertial-navigation system having an inertial measurement unit and a co-mounted driven reference module comprising a reference body of ordered medium, an alignment-field source, a coupling structure and drive source, a force transducer, and a reaction frame, the method comprising: initializing position, velocity, and attitude from an external fix or from stationary leveling; loading an empirical observation model having a provenance record and a validity envelope from a memory; chopping or stepping an oscillating drive applied to the reference body at a located precession resonance and extracting a synchronous component and an unmodulated component of the force transducer's signal; classifying a motion state from gyroscope and accelerometer statistics; executing a drive-off zero-check state and an orientation-reversal state of the driven reference module; performing an observability test on the observation model under the classified motion state; estimating an additive accelerometer bias separately from an accelerometer scale factor and from a local coupling-response state for states the observability test marks observable; correcting an inertial state of the inertial measurement unit for the estimated bias and scale factor; propagating an uncertainty of the corrected inertial state; reporting a valid, degraded, or invalid navigation state from the propagated uncertainty, a characterized update interval, and the validity envelope; and continuing to output the corrected inertial state without contemporaneous satellite-navigation input for a characterized interval and error envelope.
31. The method of claim 30, wherein the orientation-reversal state comprises rotating the driven reference module or the co-mounted inertial measurement unit and driven reference module about a horizontal axis during a stationary or quasi-static interval so that a specific-force term reverses sign while the additive accelerometer bias and a force-channel bias do not, and the estimating comprises separating the additive accelerometer bias from the reversed specific-force term.
32. The method of claim 30, wherein the local coupling-response state is reported only when the observability test marks it separable and predetermined validity criteria comprising resonance contrast, field-gradient exclusion, and thermal exclusion are satisfied, and is otherwise withheld.
33. A POAMS-native channel-partition analyzer according to claim 1, wherein the processor is further configured to report, as a prophetic native output distinct from the calibrated response class, an allocation between intrinsic-turn and orbital-winding channels inferred from the preregistered sense and mode sequence, and to report a null or unresolved allocation when the native gates fail, without requiring that the allocation be expressed through an extant force-transfer mechanism.
34. An active coupling survey system according to claim 17, wherein the reported native quantity is a change in the local angular-momentum constraint ledger of the driven reference, with conventional G, g, density, and Newtonian source parameters provided only as optional tagged comparison outputs.

ABSTRACT

Instruments and systems built on a driven ordered-medium module—an aligned medium, an alignment-field source, a resonance-locked oscillating drive, a force transducer on a defined body load path, a separate reaction load path, synchronized acquisition, a controller, and a calibration memory holding an empirical response map with provenance and validity envelope. Four products and their operating methods: a channel-response analyzer that toggles the drive, reverses alignment sense, decomposes the synchronous support-force response into sense-odd and sense-even components, and reports a response class or unclassified state; a programmable, sign-reversible microforce source calibrated against a separate force reference for force-sensor calibration; an active coupling survey and gradiometry system reporting a calibrated coupling-response coefficient with conditional gravimetric interpretation; and a coupling-referenced inertial-navigation system whose estimator separates accelerometer bias from scale factor and local coupling response with valid, degraded, and invalid states.

Driven-medium module — common product architecture (500)



Body load path (511) and reaction load path (513) are separate and both instrumented; the frame never rests on the force transducer.

FIG. 1

Driven-medium module — ordered medium in a specimen holder on a force transducer (body load path); alignment-field source and coupling structure carried by a reaction frame with its own instrumented reaction load path; drive chain, environmental sensors, synchronized acquisition, controller, and calibration memory.

Layered scope — from raw observable to operating system

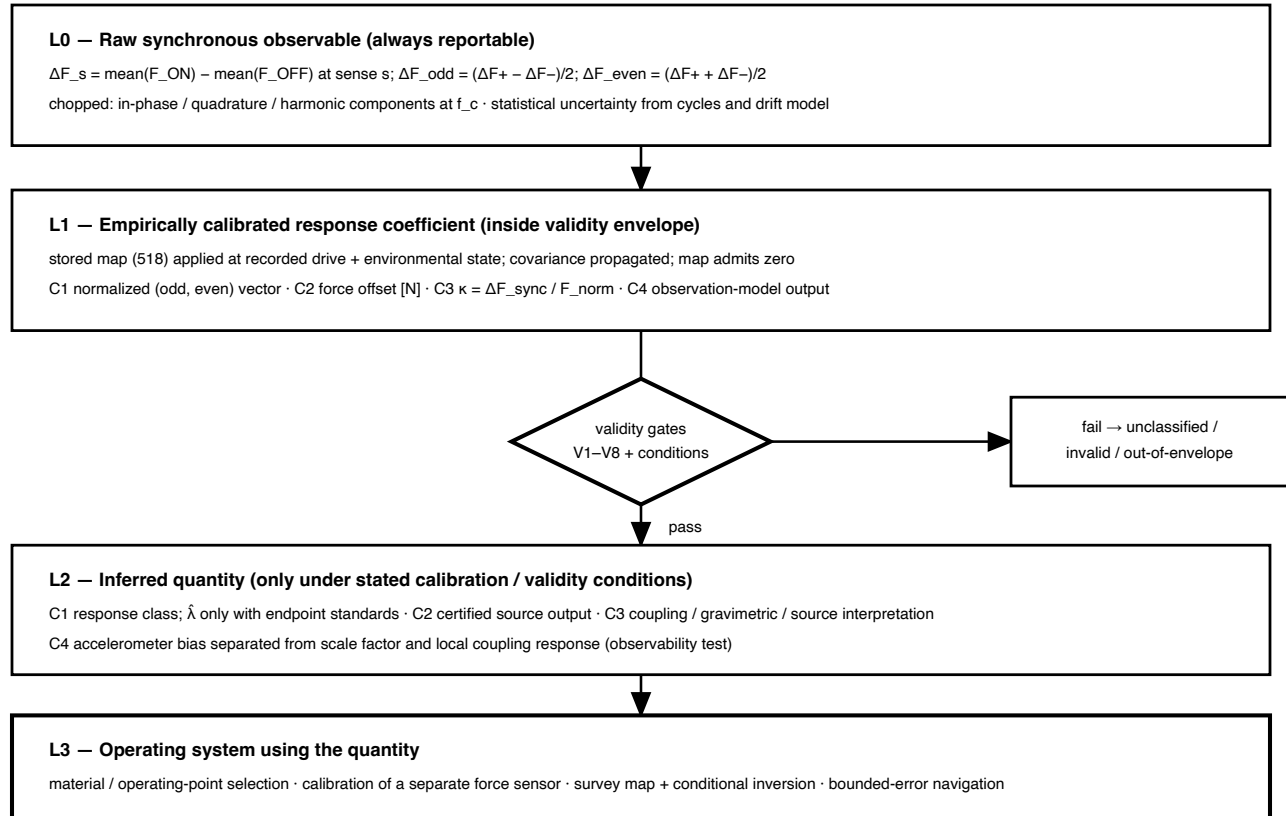
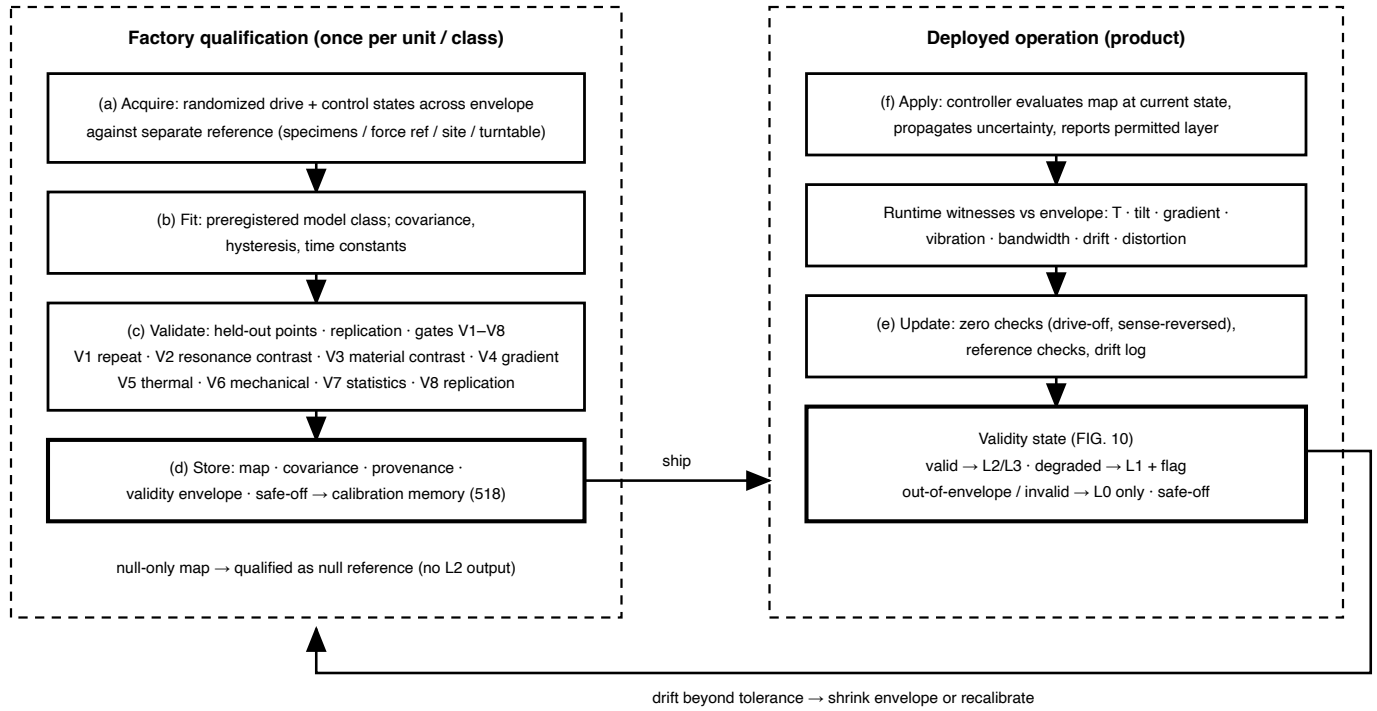


FIG. 2

Layered scope — every product output is disclosed at four layers; the validity gates sit between the calibrated coefficient (L1) and any inferred quantity (L2); failure reports unclassified, invalid, or out-of-envelope rather than a forced value.

Qualification, empirical map, deployed operation



Three tiers: (i) product + deployed method lead · (ii) qualification/calibration establish the map · (iii) validation gates govern reporting

A purchaser operates a qualified module from its stored map without repeating the qualification laboratory.

FIG. 3

Qualification and validation flow — the empirical-map procedure (acquire, fit, validate, store) performed at factory qualification; deployed operation applies the map under runtime witnesses, zero checks, and the validity state machine, with recalibration on drift.

C1 — Channel-response analyzer (550)

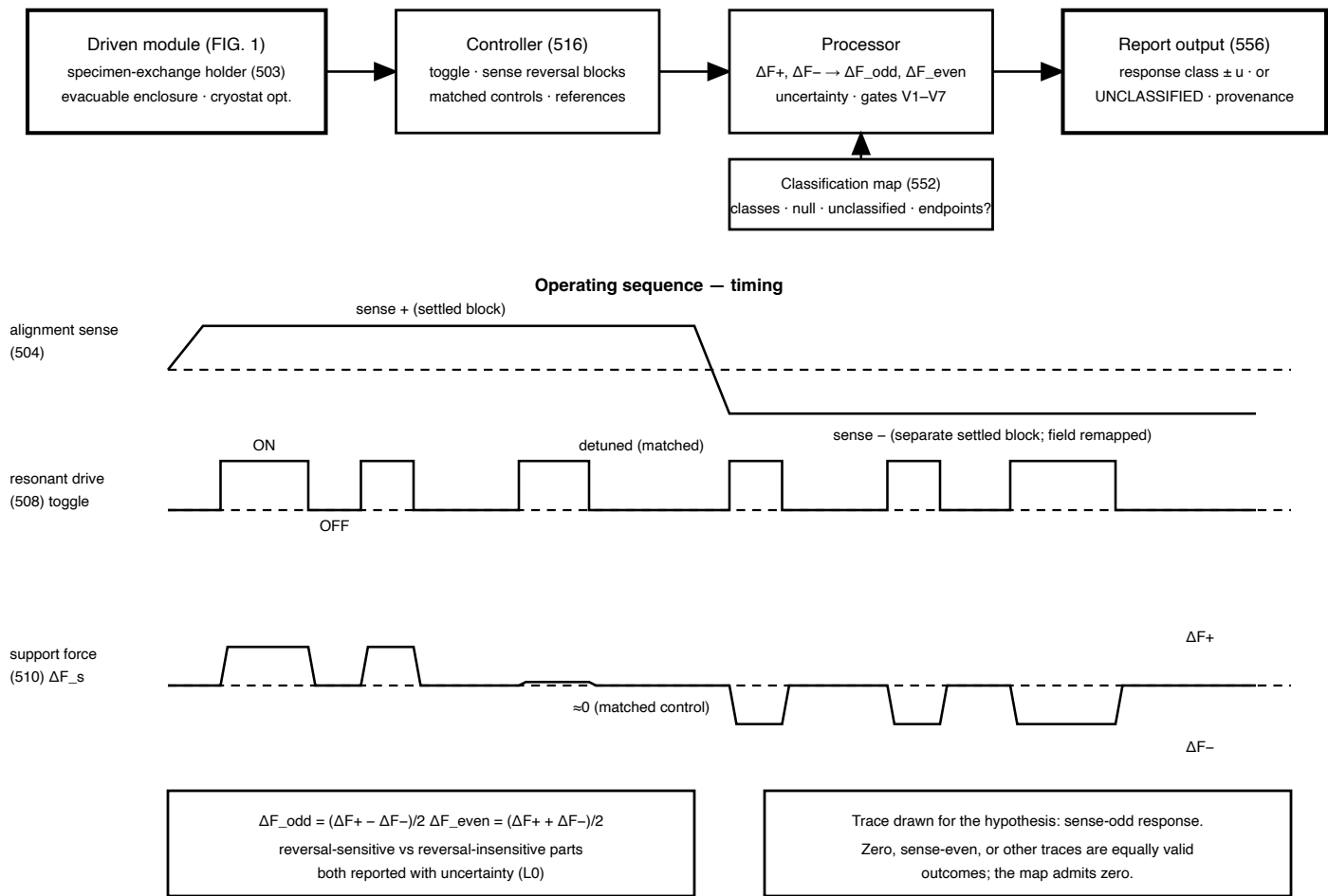


FIG. 4

C1 analyzer — product blocks (module, controller, processor, classification map, report output) and the toggle/reversal timing: drive toggled at fixed alignment sense, absorbed-energy-matched detuned control interleaved, sense reversed in a separate settled block, synchronous differential force decomposed into sense-odd and sense-even components.

C1 — Classification map (552)

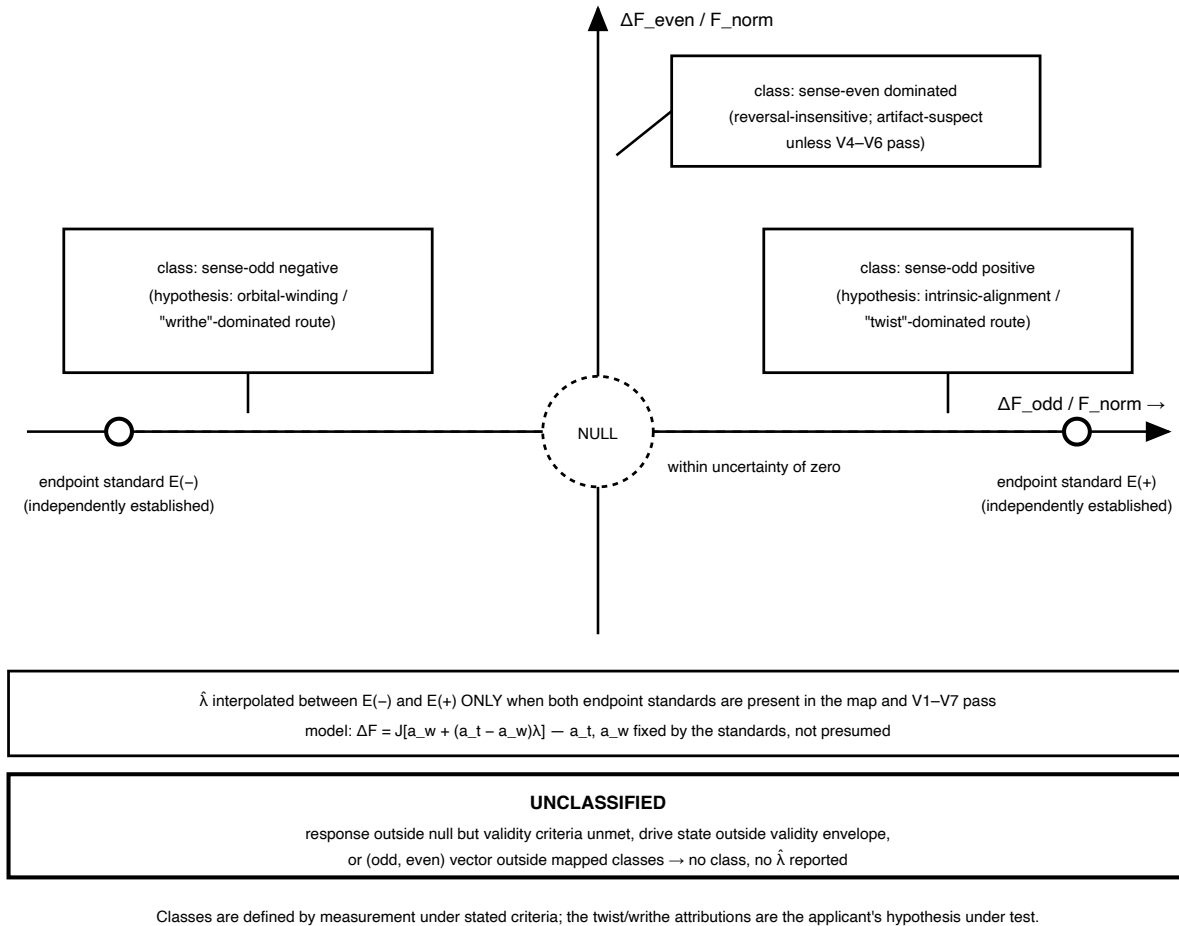
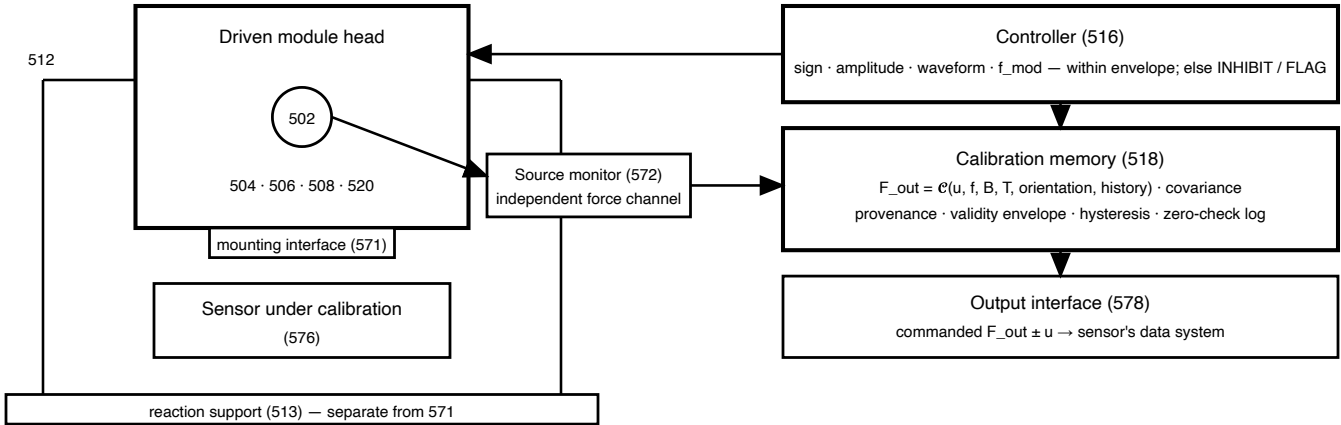


FIG. 5

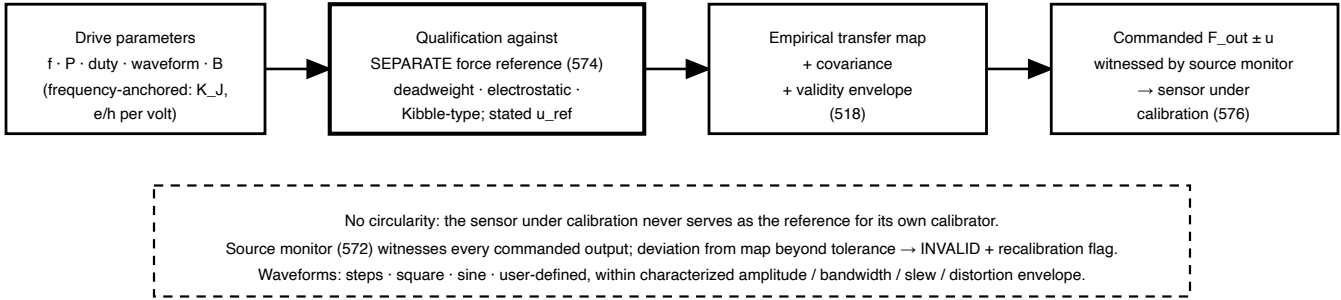
C1 classification map — null, sense-odd positive, sense-odd negative, and sense-even-dominated response classes in the normalized (odd, even) plane; endpoint standards E(–), E(+) and the condition under which a channel-partition estimate $\hat{\lambda}$ is reported; the unclassified state.

C2 — Programmable microforce source and force-sensor calibrator (570)

Deployed: source on sensor under calibration



Traceability chain (force anchored by a separate reference, not by drive electronics)



Whether a given module produces a nonzero mapped offset is established by its qualification (FIG. 3), not assumed.

FIG. 6

C2 programmable force source — driven module head transmitting body support force through a mounting interface to the sensor under calibration while frame reaction loads pass to a separate support; independent source monitor; controller and calibration memory; the corrected traceability chain anchored by qualification against a separate force reference.

C3 — Active coupling survey system (590)

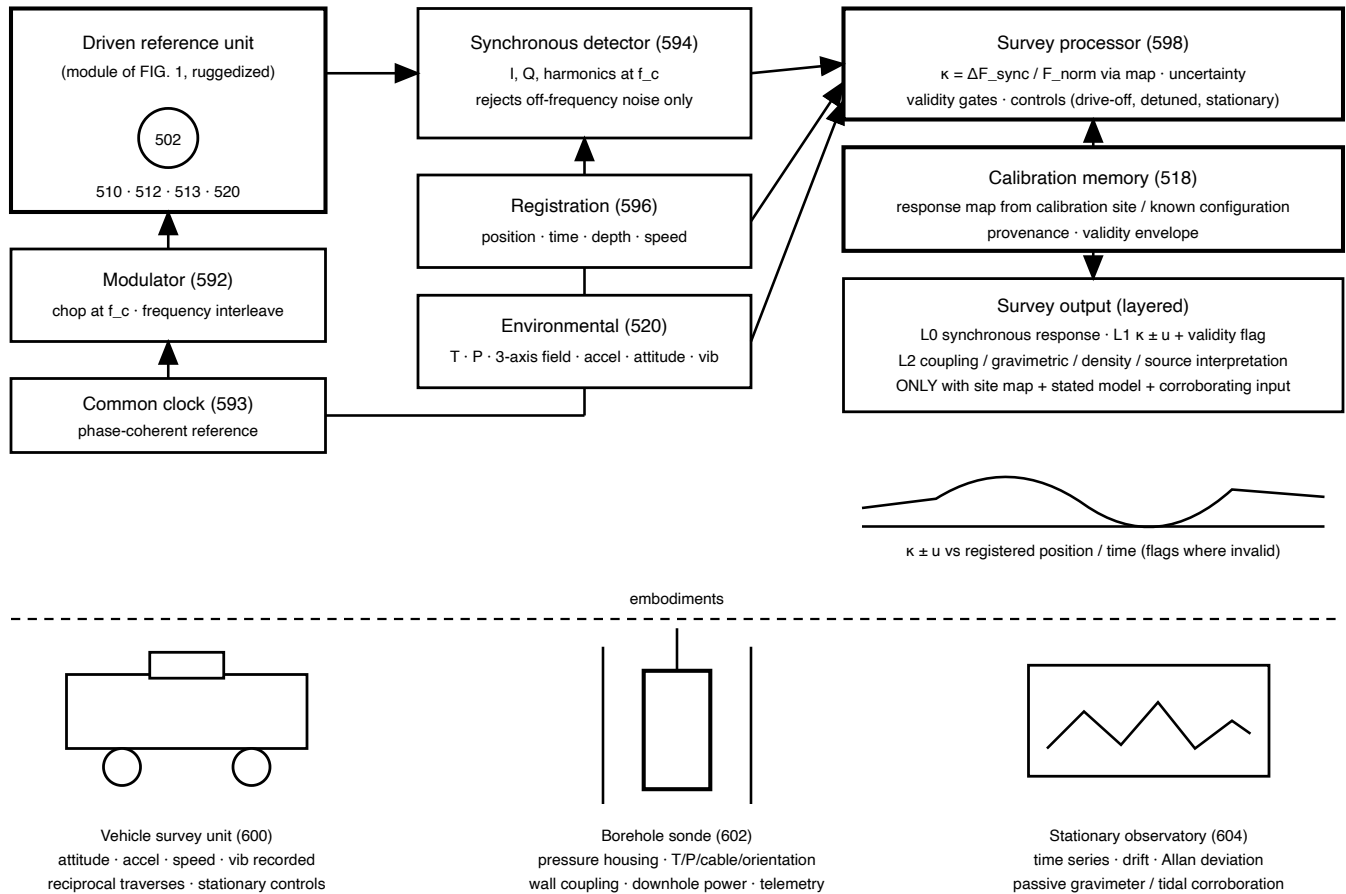


FIG. 7

C3 survey system — driven reference unit chopped at f_c under a common clock, synchronous detection, position/time registration, environmental witnesses, calibration memory, and a survey processor producing layered outputs; vehicle, borehole, and observatory embodiments.

C3 — Gradiometer: baseline projection and inversion conditions

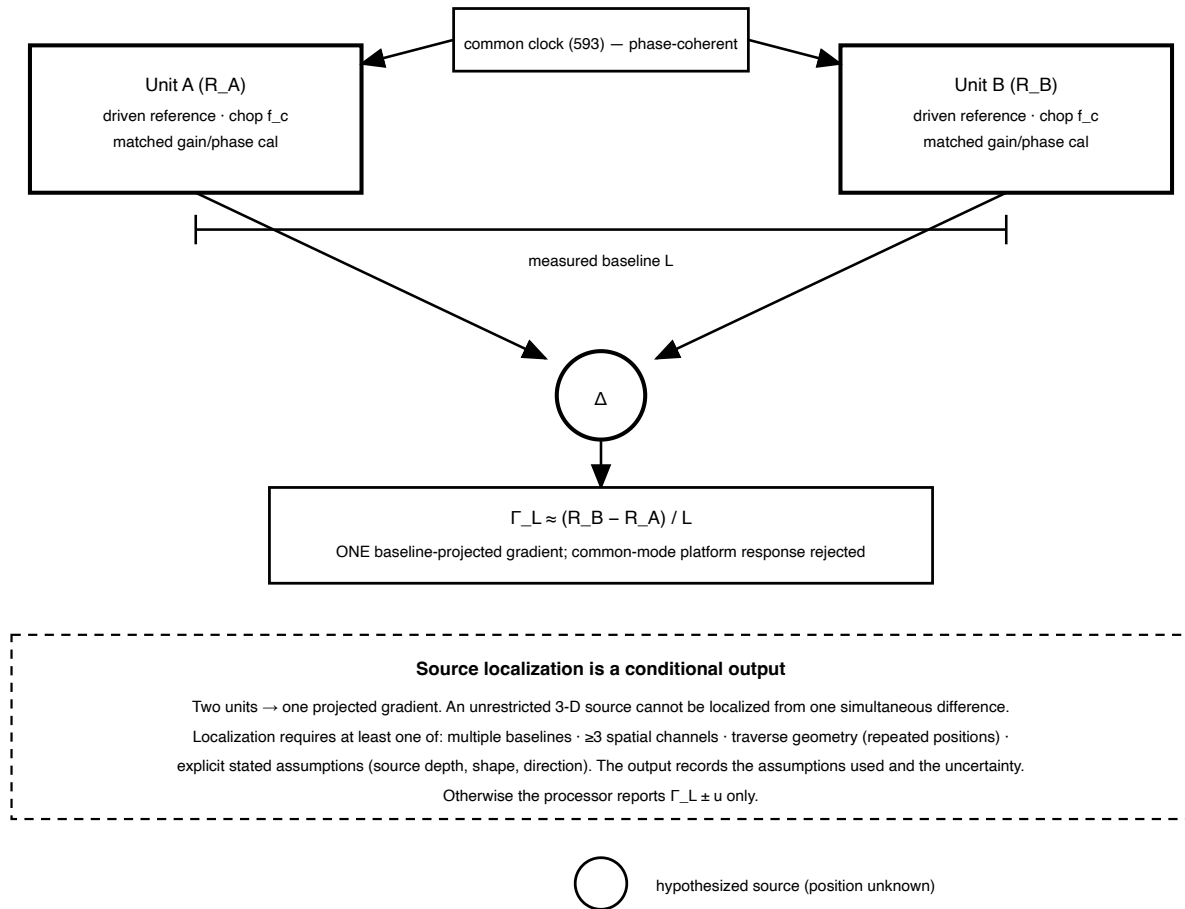


FIG. 8

C3 gradiometer — two or more driven reference units on a measured baseline with phase-coherent chopping and matched calibration yield one baseline-projected gradient; source localization is reported only when multiple baselines, additional spatial channels, traverse geometry, or stated assumptions render the inversion determinate.

C4 — Coupling-referenced inertial-navigation system (620)

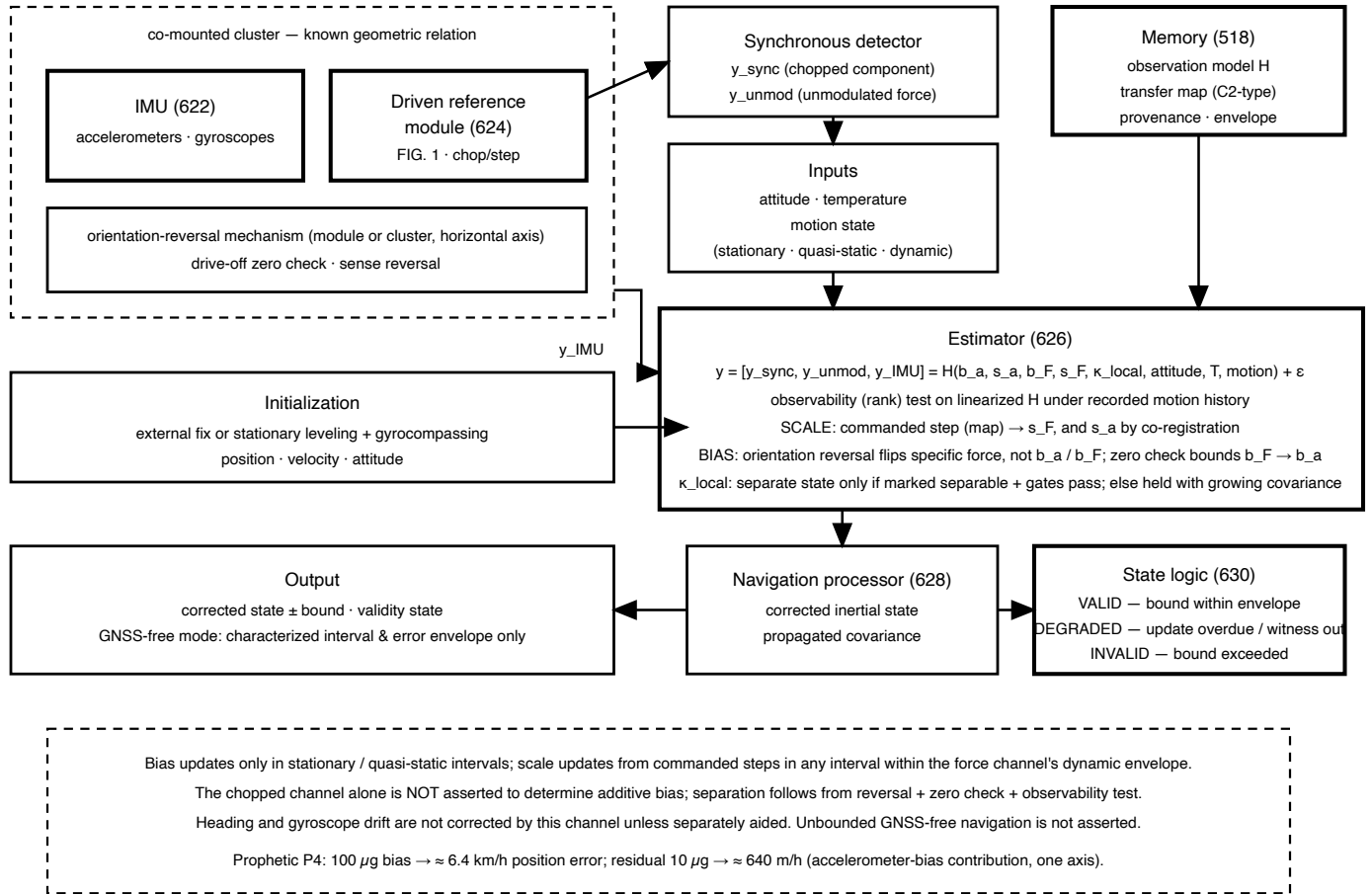
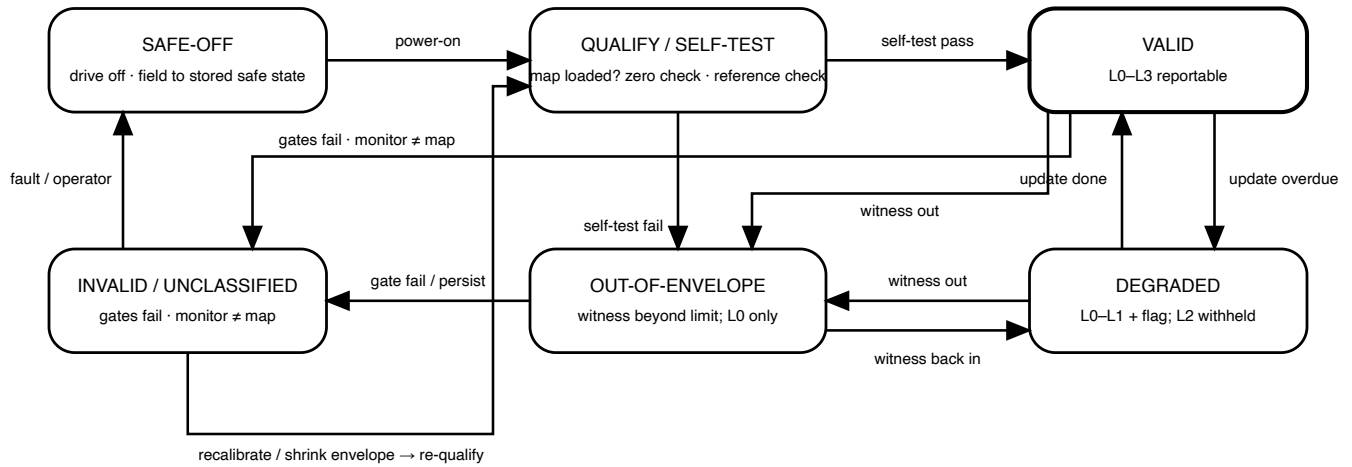


FIG. 9

C4 navigation system — IMU and driven reference module co-mounted with an orientation-reversal mechanism; synchronous detector; stored observation model; estimator separating accelerometer bias from scale factor and local coupling response under an observability test; navigation processor and valid/degraded/invalid state logic.

Controller state machine — common to C1–C4 (516)



Reporting permitted per state

VALID — raw observable, calibrated coefficient, inferred quantity (class / $\hat{\lambda}$ / F_out / κ interpretation / bias), operational use

DEGRADED — raw observable and calibrated coefficient with flag; inferred quantity withheld until required update completes

OUT-OF-ENVELOPE — raw observable only; record flagged; command inhibited (C2) or held with growing covariance (C4)

INVALID / UNCLASSIFIED — no class, no certified output, no inferred coefficient; residuals logged; recalibration flag

SAFE-OFF — drive off; alignment field to stored safe state; last valid record preserved

Product-specific names: C1 UNCLASSIFIED · C2 INHIBIT/FLAG · C3 validity flag per sample · C4 VALID/DEGRADED/INVALID navigation state

FIG. 10

Controller state machine common to the four products — safe-off, qualify/self-test, valid, degraded, out-of-envelope, and invalid/unclassified states with the transitions and the reporting layer permitted in each state.